

11767 HORROBIN FARM, TURTON

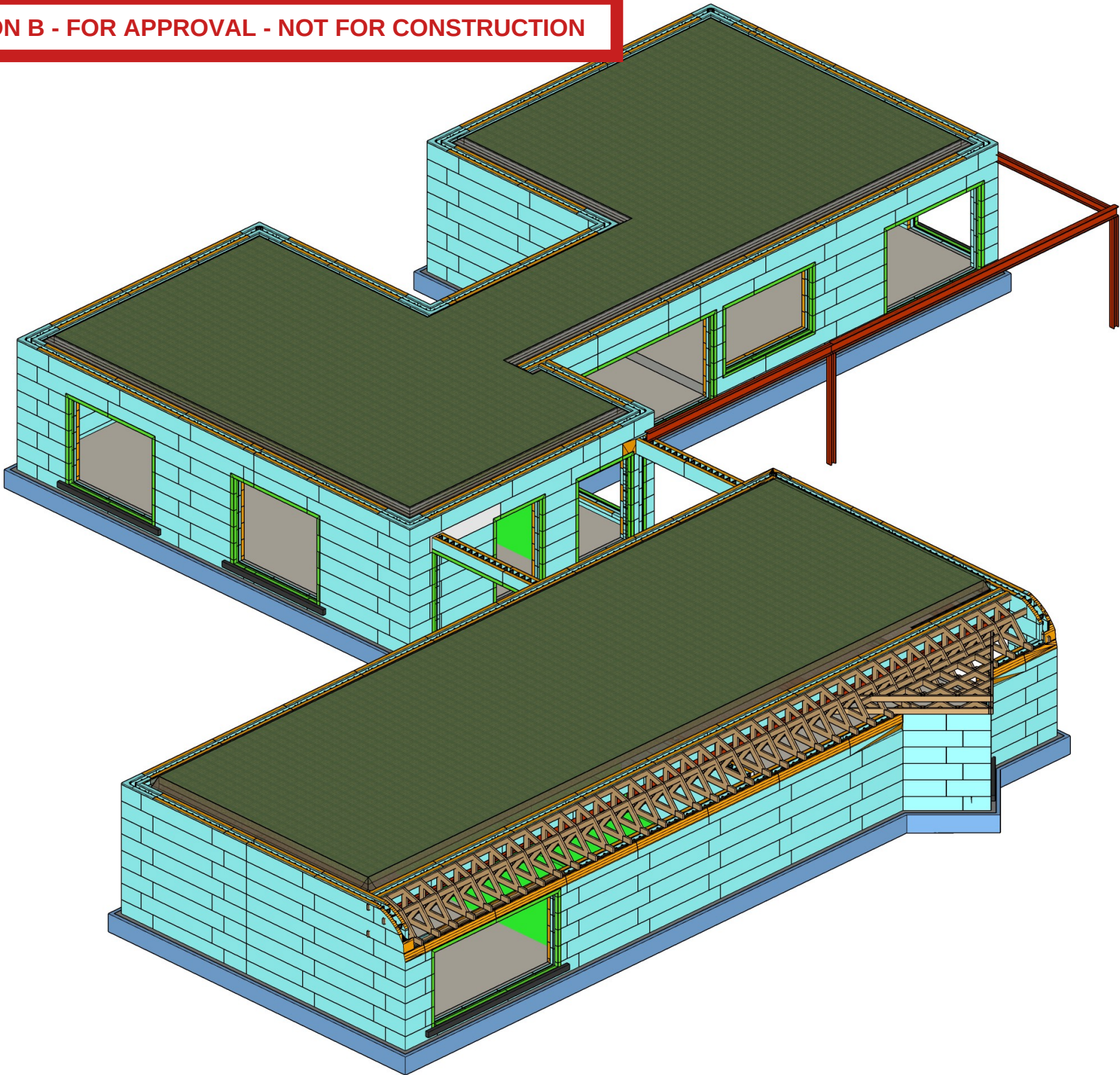
UKPCS-002

ICF APPROVAL DRAWING PACK | OPTION B: ICF 100MM IN FROM SLAB EDGE

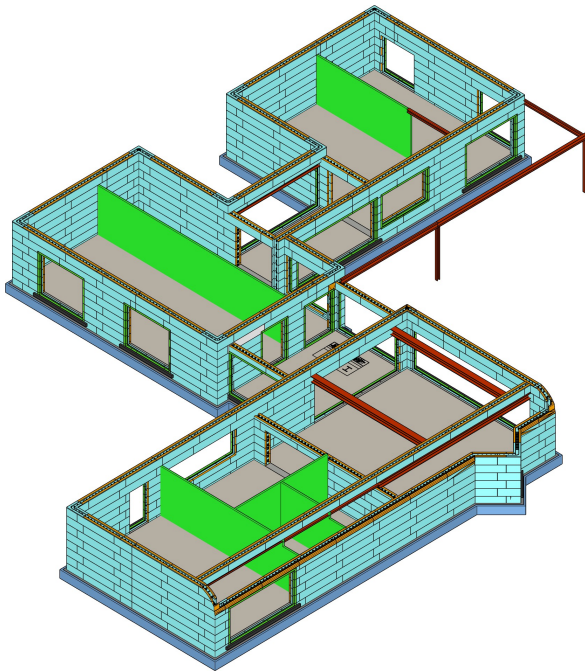
Issue P01 | 15.09.26

Client: Mr A Kirkpatrick

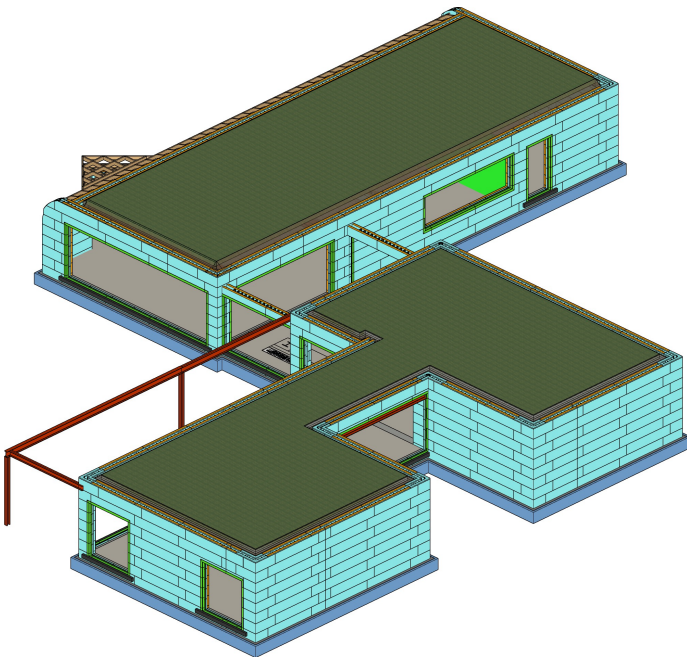
OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION



NUDURA ICF STRUCTURE WITH ROOF, STEEL AND GROUND FLOOR BUILD-UP, FROM THE SOUTH-EAST



ICF STRUCTURE, ROOF BUILD-UP REMOVED



VIEW FROM THE NORTH-WEST



OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION

11767 HORROBIN FARM, TURTON

ICF APPROVAL DRAWING PACK | OPTION B: ICF 100MM IN FROM SLAB EDGE

Client: Mr A Kirkpatrick | Issue P01 15.09.26 | Nudura ICF with BuildEase

DRAWING No.	TITLE	SCALE	SIZE	REV	PAGE
GENERAL ARRANGEMENT					
UKPCS-002-GA-01B	Ground Floor ICF General Arrangement	1:150	A3	P01	3
ELEVATIONS					
UKPCS-002-EL-01B	North and South Elevations	1:125	A3	P01	4
UKPCS-002-EL-02B	East and West Elevations	1:150	A3	P01	5
SECTIONS					
UKPCS-002-SE-01B	Section A-A	1:75	A3	P01	6
UKPCS-002-SE-02B	Section B-B	1:50	A3	P01	7
UKPCS-002-SE-03B	Section C-C	1:75	A3	P01	8
UKPCS-002-SE-04B	Section D-D	1:75	A3	P01	9
UKPCS-002-SE-05B	Section E-E	1:75	A3	P01	10
DETAILS					
UKPCS-002-DT-01B	Threshold Detail - External	1:10	A3	P01	11
UKPCS-002-DT-02B	Threshold Detail - Internal	1:10	A3	P01	12
ENGINEER					
UKPCS-002-EQ-01B	Questions for the Engineer	NTS	A3	P01	13

PACK REGISTER

ISSUE	
Item	Value
Status	FOR APPROVAL
Revision	P01
Sheets	11
Pages (incl. cover + contents)	13

- NOTES
- Click any row to open that sheet (page numbers count the cover as page 1). On every sheet, click the drawing number (bottom right) to come back to this contents page.
 - On GA-01 the red EQ tags open their question on EQ-01 and the section heads open the section sheets.
 - Approval issue P01: for comment and approval, not for construction. Levels above GF FFL (143.750 AOD). Structural engineer's drawings take precedence for structure.

REVISIONS		
REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE			
APPROVAL PACK CONTENTS			
Drawing register with links			
SCALE	SIZE	DATE	DRAWN
NTS	A3	15.09.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-CT-00B	P01



ENGINEER'S LINTELS (ICF OPENINGS)

Mark	HxW	Reinforcement
L-1	300x150	2-H10 top, 2-H12 bott., H8 links @ 200 c/c, RC30
L-2	300x150	1-H10 top, 1-H12 bott., H8 links @ 200 c/c, RC30
L-3	300x150	1-H10 top, 1-H12 bott., H8 links @ 200 c/c, RC30
L-4	450x150	2-H12 top, 2-H20 bott., H8 links @ 200 c/c, RC30
L-5	450x150	2-H12 top, 2-H16 bott., H8 links @ 200 c/c, RC30
L-6	450x150	2-H12 top, 2-H20 + 2-H16 bott., H10 full links @ 200 c/c, RC30 (or 305x165x40 UB S355)
L-7	600x150	2-H12 top, 2-H20 + 2-H16 bott., H8 links @ 200 c/c, RC30
L-8	300x150	2-H10 top, 2-H12 bott., H8 links @ 200 c/c, RC30
L-9	300x150	2-H12 top, 2-H16 bott., H8 links @ 200 c/c, RC30
L-10	300x150	2-H12 top, 2-H16 bott., H8 links @ 200 c/c, RC30
L-11	300x150	2-H10 top, 2-H12 bott., H8 links @ 200 c/c, RC30
L-12	450x150	2-H12 top, 2-H16 bott., H8 links @ 200 c/c, RC30
L-13	600x150	2-H12 top, 2-H20 bott., H8 links @ 200 c/c, RC30 (option 2: 450x150, 2-H12 top, 2-H20 + 2-H16 bott., H10 links)
L-14	300x150	2-H10 top, 2-H12 bott., H8 links @ 200 c/c, RC30
L-15	300x150	2-H10 top, 2-H12 bott., H8 links @ 200 c/c, RC30
L-16	300x150	2-H12 top, 2-H16 bott., H8 links @ 200 c/c, RC30
L-17	300x150	2-H12 top, 2-H20 bott., H8 links @ 200 c/c, RC30

From Carb Zero 26-92-01 rev C lintel schedule.

OPENING SCHEDULE

Ref	Width	Height	Sill	Head	Lintel
CW01, D01, CW02	4098	2400	0	+2400	L-16
CW03	4881	1500	+900	+2400	L-4
CW04 (1)	1338	2140	-20	+2120	L-2
CW05	4099	2400	0	+2400	L-17
CW06	2900	2200	+200	+2400	L-9
CW07	2987	1800	+600	+2400	L-14
CW08	3250	2400	0	+2400	L-14
CW09	2200	2400	0	+2400	L-15
CW10	2200	2200	0	+2200	L-15
CW13	2875	2400	0	+2400	L-10
CW14	2875	2400	0	+2400	L-10
D1 (3)	5775	2400	0	+2400	B-6 over
D2	5125	2400	0	+2400	L-13
D02	1250	2400	0	+2400	L-3
D3	7855	2400	0	+2400	L-7
D4	5675	2400	0	+2400	L-6
D13	1587	2400	0	+2400	L-8
D14	1800	2400	0	+2400	L-8
D15	4250	2060	0	+2060	L-1
VOID W19 (2)	4143	2070	0	+2070	L-5
VOID W4	2887	2400	0	+2400	L-12
VOID W7	2400	2400	0	+2400	L-11

Structural openings in the ICF, sill and head above GF FFL. (1) CW04: head lowered from +2355 to take L-2 (300) under the +2420 wall top. (2) VOID W19: head lowered from +2120 to take L-5 (450) under the +2520 wall top. (3) D1: height = clear under B-6 (BOS +2400); ICF open to +2654 with B-6 in the head (EQ-02). VOID = subtract cut in the ICF (no end caps / lintel buck modelled).

Mark	Section (engineer)	BOS	TOS
B-1	152x152x37 UC S355 with 10mm top plate	+2698	+2860
B-2	254x146x31 UB S355 with 10mm top plate & 200x100x10 ledger angle one side	+2609	+2860
B-3	305x305x158 UC S355 with 200x100x10 ledger angles both sides	+2533	+2860
B-4	305x305x118 UC S355 with 200x100x10 ledger angles both sides	+2546	+2860
B-5	305x305x198 UC S355 with 200x100x10 ledger angle one side	+2520	+2860
B-6	254x254x73 UC S355 with 200x100x10 ledger angle one side	+2400	+2654
B-7	152x152x37 UC S355 with 10mm bottom plate both sides	+2688	+2850
B-8	254x146x43 UB S355	+2800	+3060
B-9	254x146x31 UB S355	+2800	+3051
TIE	203x133x25 UB S355	+2848	+3051
C-1	152x152x23 UC S355	-280	+2800
C-1	152x152x23 UC S355	-280	+2800

BOS / TOS = bottom / top of steel above GF FFL (C-1 columns: base / top).

1. ICF: Nudura 6" XR35, 101.6 EPS + 152.4 core + 101.6 EPS = 355.6. Dimensions to the ICF faces.
2. Levels above GF FFL (143.750 AOD). Sill / head = structural opening in the ICF.
3. EXT ICF150; 150mm core ICF walls, concrete grade C30, 10mm aggregate; polypropylene fibre reinforcement, dosage 2.5kg/m³; H16 dowel bars between lifts @ max 450mm c/c, 800mm long with 200mm embedment, resin fixed.
4. Lintel (L-n) and steel (B-n, C-1) marks from Carb Zero 26-92-01 rev C, which takes precedence. Red EQ tags = queries to the engineer on UKPCS-002-EQ-01B (click to open).
5. WS: wall starters between ICF & block wall. Frame cramps @ max 225mm vert. c/c. Masonry walls shown for context.
6. Steel over dashed. BOS / TOS = bottom / top of steel above GF FFL.

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



PROJECT	11767 Horrobin Farm, Turton
---------	-----------------------------

GROUND FLOOR ICF GENERAL ARRANGEMENT

ICF walls, openings, lintels, steel over and engineer's notes

SCALE 1:150	SIZE A3	DATE 15.09.26	DRAWN TS
----------------	------------	------------------	-------------

DRAWING No.

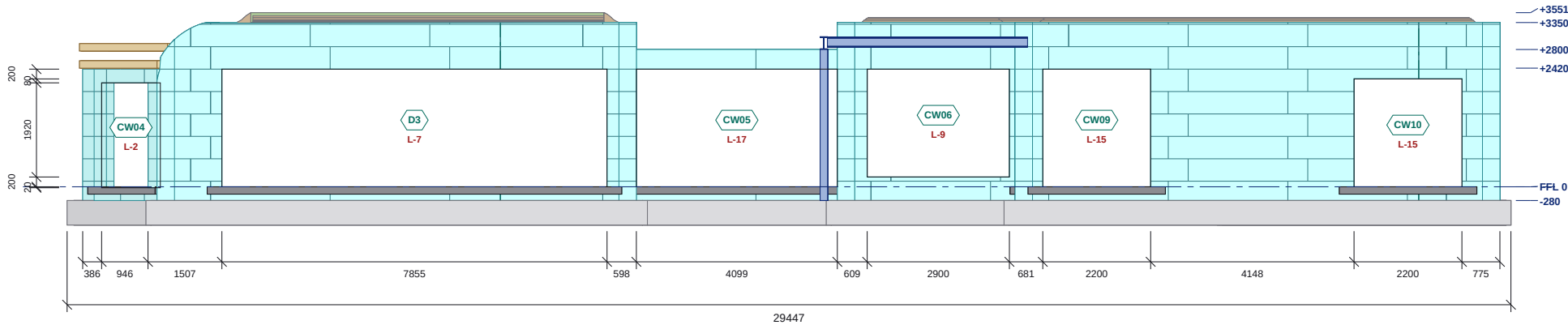
UKPCS-002-GA-01B

P01

-  EPS insulation (Nudura form)
-  Concrete core
-  End cap at opening
-  Load-bearing masonry wall (context)
-  Steel over (dashed), marked
-  Steel column cut
-  Engineer's pier bars (n-H12 vertical bars in ICF pier) / WS wall starters
-  Setting-out check diagonal (external ICF corners)
-  Section line (see SE sheet)

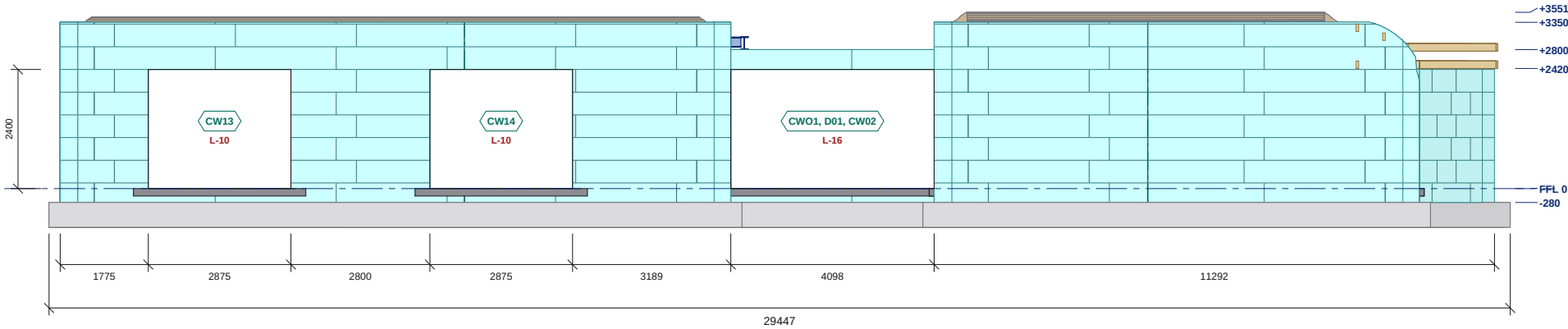
GROUND FLOOR ICF PLAN

Scale 1:150 @ A3 | Plan cut at 1200 above GF FFL



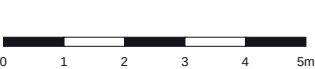
NORTH ELEVATION

Scale 1:125 @ A3 | Levels above GF FFL (AOD in the schedule); left: sills and heads on this face



SOUTH ELEVATION

Scale 1:125 @ A3 | Levels above GF FFL (AOD in the schedule); left: sills and heads on this face



- ICF external EPS face
- Block course joint
- Stack joint
- Steel
- Roof build-up (slab / earth / sedum)
- Curved trusses / timbers
- GF slab + foundation build-up, ACO

ELEVATION SCHEDULES

NORTH ELEVATION: OPENINGS

Ref	Width	Height	Sill	Head	Lintel
CW04 (1)	1338	2140	-20	+2120	L-2
CW05	4099	2400	0	+2400	L-17
CW06	2900	2200	+200	+2400	L-9
CW09	2200	2400	0	+2400	L-15
CW10	2200	2200	0	+2200	L-15
D3	7855	2400	0	+2400	L-7

Structural openings in the ICF; sill and head above GF FFL. (1) CW04: head lowered from +2355 to take L-2 (300) under the +2420 wall top.

SOUTH ELEVATION: OPENINGS

Ref	Width	Height	Sill	Head	Lintel
CW01, D01, CW02	4098	2400	0	+2400	L-16
CW13	2875	2400	0	+2400	L-10
CW14	2875	2400	0	+2400	L-10

Structural openings in the ICF; sill and head above GF FFL.

NORTH ELEVATION: LEVELS

Level	FFL	AOD
underside of ICF	-280	143.470
GF FFL	0	143.750
top of ICF	+2420	146.170
top of ICF	+2800	146.550
top of ICF	+3350	147.100
top of roof build-up	+3551	147.301

SOUTH ELEVATION: LEVELS

Level	FFL	AOD
underside of ICF	-280	143.470
GF FFL	0	143.750
top of ICF	+2420	146.170
top of ICF	+2800	146.550
top of ICF	+3350	147.100
top of roof build-up	+3551	147.301

NOTES

- ICF: Nudura 6" XR35 (101.6 EPS + 152.4 core + 101.6 EPS = 355.6). External EPS face shown with the block course joints, vertical block joints and stack joints as populated.
- Levels above GF FFL (143.750 AOD). Openings are structural openings in the ICF; glazing, doors, render and finishes by others, not shown.
- Roof build-up, steel, curved trusses, GF slab / foundation build-up (BR11) and ACO drains shown as modelled.
- See UKPCS-002-GA-01B for plan setting out, lintels and steel.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE

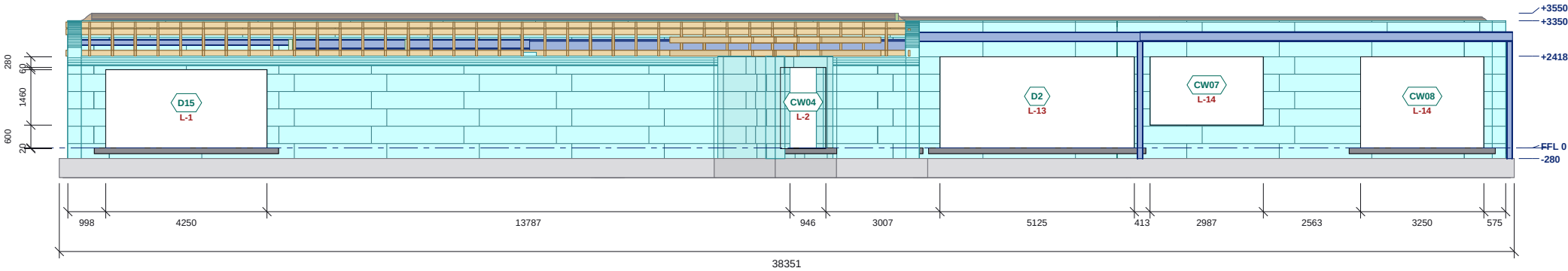
NORTH AND SOUTH ELEVATIONS

ICF elevations with openings, levels and roof build-up

SCALE	SIZE	DATE	DRAWN
1:125	A3	15.09.26	TS

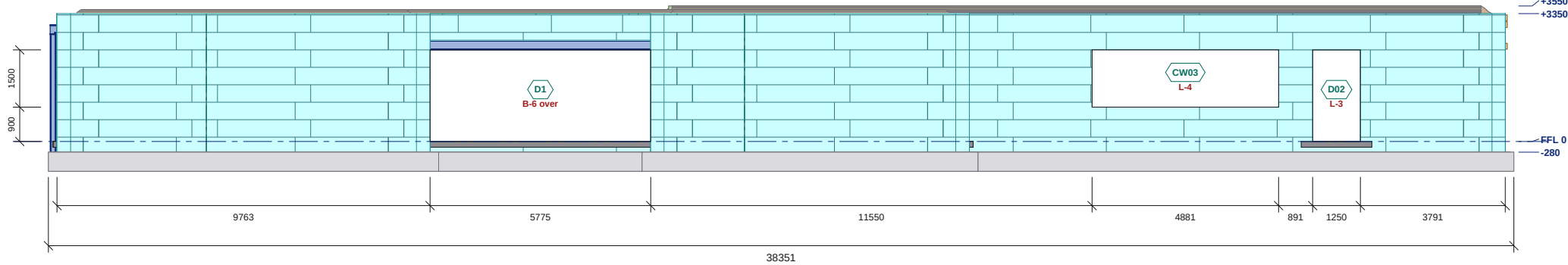
STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-EL-01B	P01

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION



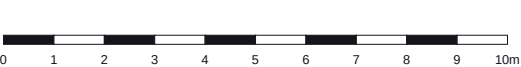
EAST ELEVATION

Scale 1:150 @ A3 | Levels above GF FFL (AOD in the schedule); left: sills and heads on this face



WEST ELEVATION

Scale 1:150 @ A3 | Levels above GF FFL (AOD in the schedule); left: sills and heads on this face



- ICF external EPS face
- Block course joint
- Stack joint
- Steel
- Roof build-up (slab / earth / sedum)
- Curved trusses / timbers
- GF slab + foundation build-up, ACO

ELEVATION SCHEDULES

EAST ELEVATION: OPENINGS

Ref	Width	Height	Sill	Head	Lintel
CW04 (1)	1338	2140	-20	+2120	L-2
CW07	2987	1800	+600	+2400	L-14
CW08	3250	2400	0	+2400	L-14
D2	5125	2400	0	+2400	L-13
D15	4250	2060	0	+2060	L-1

Structural openings in the ICF; sill and head above GF FFL. (1) CW04: head lowered from +2355 to take L-2 (300) under the +2420 wall top.

WEST ELEVATION: OPENINGS

Ref	Width	Height	Sill	Head	Lintel
CW03	4881	1500	+900	+2400	L-4
D1 (3)	5775	2400	0	+2400	B-6 over
D02	1250	2400	0	+2400	L-3

Structural openings in the ICF; sill and head above GF FFL. (3) D1: height = clear under B-6 (BOS +2400); ICF open to +2654 with B-6 in the head (EQ-02).

EAST ELEVATION: LEVELS

Level	FFL	AOD
underside of ICF	-280	143.470
GF FFL	0	143.750
top of ICF	+2418	146.168
top of ICF	+3350	147.100
top of roof build-up	+3550	147.300

WEST ELEVATION: LEVELS

Level	FFL	AOD
underside of ICF	-280	143.470
GF FFL	0	143.750
top of ICF	+3350	147.100
top of roof build-up	+3550	147.300

NOTES

- ICF: Nudura 6" XR35 (101.6 EPS + 152.4 core + 101.6 EPS = 355.6). External EPS face shown with the block course joints, vertical block joints and stack joints as populated.
- Levels above GF FFL (143.750 AOD). Openings are structural openings in the ICF; glazing, doors, render and finishes by others, not shown.
- Roof build-up, steel, curved trusses, GF slab / foundation build-up (BR11) and ACO drains shown as modelled.
- See UKPCS-002-GA-01B for plan setting out, lintels and steel.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE

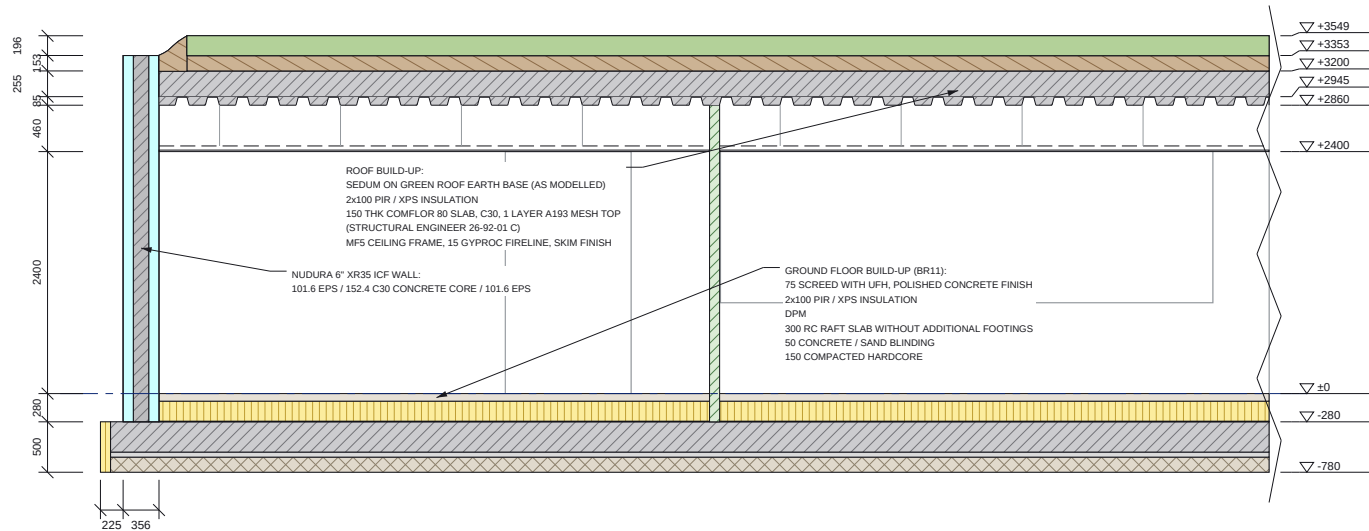
EAST AND WEST ELEVATIONS

ICF elevations with openings, levels and roof build-up

SCALE	SIZE	DATE	DRAWN
1:150	A3	15.09.26	TS

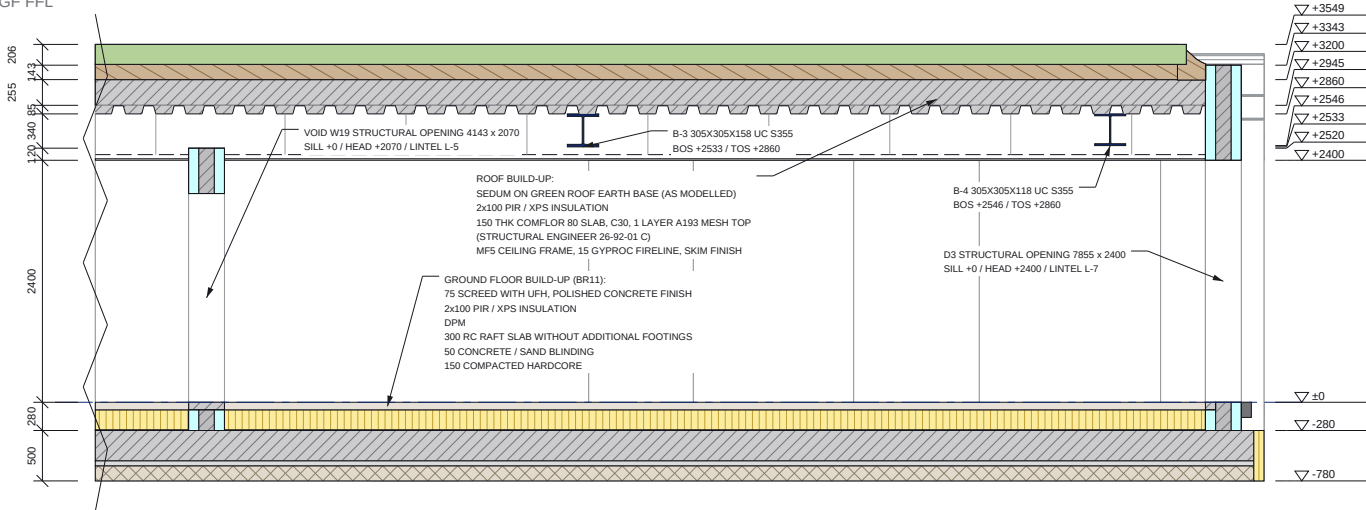
STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-EL-02B	P01

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION



SECTION A-A (PART 1 OF 2)

Scale 1:75 @ A3 | Looking west; levels above GF FFL



SECTION A-A (PART 2 OF 2)

Scale 1:75 @ A3 | Looking west; levels above GF FFL

- ICF EPS (cut)

ICF concrete core / RC (cut)

Steel (cut)

Screed
- Insulation (PIR / XPS, perimeter)

Blinding / hardcore

Roof earth

Sedum
- Masonry wall (cut)

Suspended ceiling (MF5 + 15 Fireline)

Comflor 80 deck (profile line)

Beyond the cut (line work)



SECTION A-A SCHEDULES

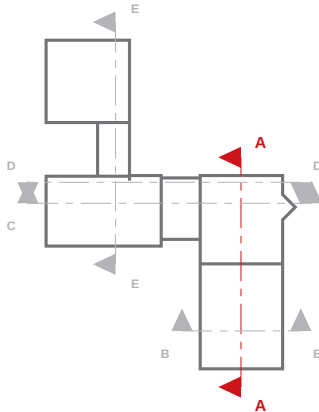
LEVELS ON THIS SECTION

Level	FFL	AOD
underside of hardcore	-780	142.970
underside of ICF / top of RC raft	-280	143.470
GF FFL	0	143.750
underside of suspended ceiling	+2400	146.150
top of ICF	+2520	146.270
B-3 BOS	+2533	146.283
B-4 BOS	+2546	146.296
underside of Comflor deck	+2860	146.610
roof slab soffit	+2945	146.695
top of roof slab	+3200	146.950
top of ICF	+3343	147.093
top of ICF	+3353	147.103
top of roof build-up	+3549	147.299

STEEL CUT BY THIS SECTION

Mark	Section (engineer)	BOS	TOS
B-3	305x305x158 UC S355 with 200x100x10 ledger angles both sides	+2533	+2860
B-4	305x305x118 UC S355 with 200x100x10 ledger angles both sides	+2546	+2860

SECTION KEY



NOTES

- Section A-A on the line of the architect's section (BR11). Cut elements hatched / filled; the model beyond the cut in line work.
- Levels above GF FFL (143.750 AOD); the level table gives AOD.
- ICF: Nudura 6" XR35, 101.6 EPS + 152.4 C30 core + 101.6 EPS = 355.6. EXT ICF150: 150mm core ICF walls, concrete grade C30, 10mm aggregate; polypropylene fibre reinforcement, dosage 2.5kg/m3; H16 dowel bars between lifts @ max 450mm c/c, 800mm long with 200mm embedment, resin fixed.
- Ground floor (BR11): 75 screed with UFH / 200 PIR-XPS / DPM / 300 RC raft slab without additional footings / 50 blinding / 150 compacted hardcore; 100 perimeter insulation outside the ICF. Thresholds: ICF cut to -75 under the screed, concrete infill, ACO drain (size assumed 100x150).
- Roof: 150 thk Comflor 80 slab, C30, 1 layer A193 mesh top (26-92-01 C); green roof build-up (earth + sedum) as modelled; waterproofing to specialist design. Timber roof over the east wall to specialist supplier design. Comflor 80 deck drawn to the architect's profile (80 deep, 300 pitch) on the bearing steel / walls; MF5 suspended ceiling, 15 Gyproc Fireline, FFL to underside 2400 (BR11-13), drawn only.
- Steel marks, BOS / TOS: Carb Zero 26-92-01 rev C. See UKPCS-002-GA-01B / EL-01B / EL-02B.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge

CLIENT

Mr A Kirkpatrick

PROJECT

11767 Horrobin Farm, Turton

DRAWING TITLE

SECTION A-A

ICF section with levels and build-ups

SCALE

1:75

SIZE

A3

DATE

15.09.26

DRAWN

TS

STATUS

FOR APPROVAL

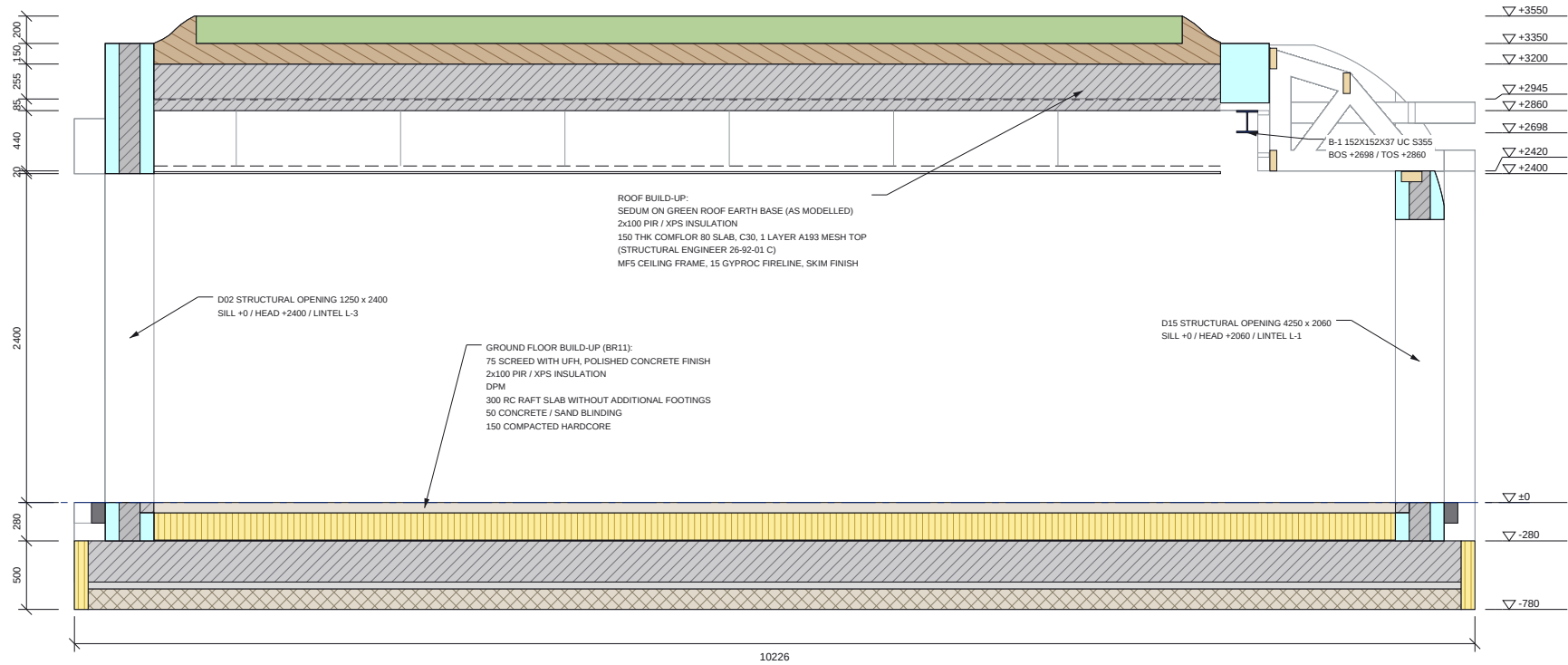
DRAWING No.

UKPCS-002-SE-01B

REV

P01

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION



SECTION B-B

Scale 1:50 @ A3 | Looking north; levels above GF FFL

	ICF EPS (cut)		Insulation (PIR / XPS, perimeter)		Masonry wall (cut)
	ICF concrete core / RC (cut)		Blinding / hardcore		Suspended ceiling (MF5 + 15 Fireline)
	Steel (cut)		Comflor 80 deck (profile line)		Beyond the cut (line work)
	Screed		Roof earth		
			Sedum		



SECTION B-B SCHEDULES

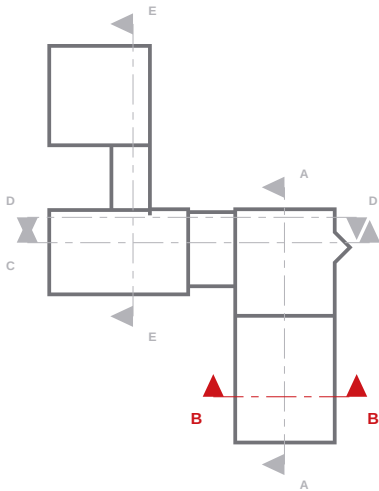
LEVELS ON THIS SECTION

Level	FFL	AOD
underside of hardcore	-780	142.970
underside of ICF / top of RC raft	-280	143.470
GF FFL	0	143.750
underside of suspended ceiling	+2400	146.150
top of ICF	+2420	146.170
B-1 BOS	+2698	146.448
underside of Comflor deck	+2860	146.610
roof slab soffit	+2945	146.695
top of roof slab	+3200	146.950
top of ICF	+3350	147.100
top of roof build-up	+3550	147.300

STEEL CUT BY THIS SECTION

Mark	Section (engineer)	BOS	TOS
B-1	152x152x37 UC S355 with 10mm top plate	+2698	+2860

SECTION KEY



NOTES

- Section B-B on the line of the architect's section (BR12). Cut elements hatched / filled; the model beyond the cut in line work.
- Levels above GF FFL (143.750 AOD); the level table gives AOD.
- ICF: Nudura 6" XR35, 101.6 EPS + 152.4 C30 core + 101.6 EPS = 355.6. EXT ICF150: 150mm core ICF walls, concrete grade C30, 10mm aggregate; polypropylene fibre reinforcement, dosage 2.5kg/m³; H16 dowel bars between lifts @ max 450mm c/c, 800mm long with 200mm embedment, resin fixed.
- Ground floor (BR11): 75 screed with UFH / 200 PIR-XPS / DPM / 300 RC raft slab without additional footings / 50 blinding / 150 compacted hardcore; 100 perimeter insulation outside the ICF. Thresholds: ICF cut to -75 under the screed, concrete infill, ACO drain (size assumed 100x150).
- Roof: 150 thk Comflor 80 slab, C30, 1 layer A193 mesh top (26-92-01 C); green roof build-up (earth + sedum) as modelled; waterproofing to specialist design. Timber roof over the east wall to specialist supplier design. Comflor 80 deck drawn to the architect's profile (80 deep, 300 pitch) on the bearing steel / walls; MF5 suspended ceiling, 15 Gyproc Fireline, FFL to underside 2400 (BR11-13), drawn only.
- Steel marks, BOS / TOS: Carb Zero 26-92-01 rev C. See UKPCS-002-GA-01B / EL-01B / EL-02B.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



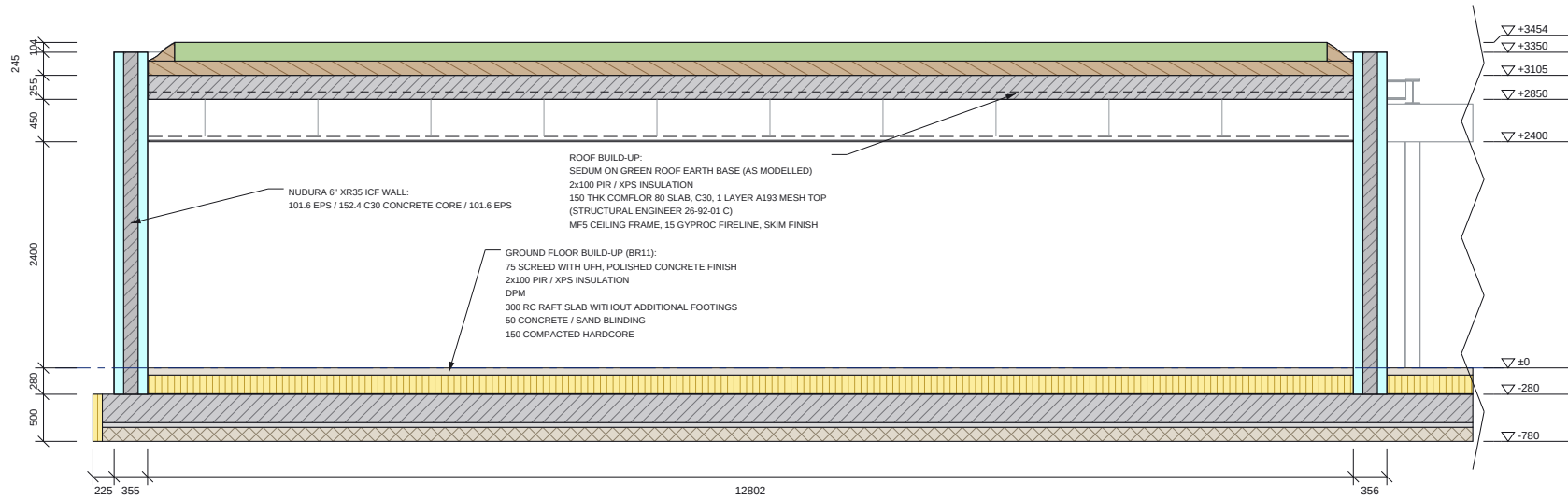
CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE			
SECTION B-B			
ICF section with levels and build-ups			

SCALE	SIZE	DATE	DRAWN
1:50	A3	15.09.26	TS

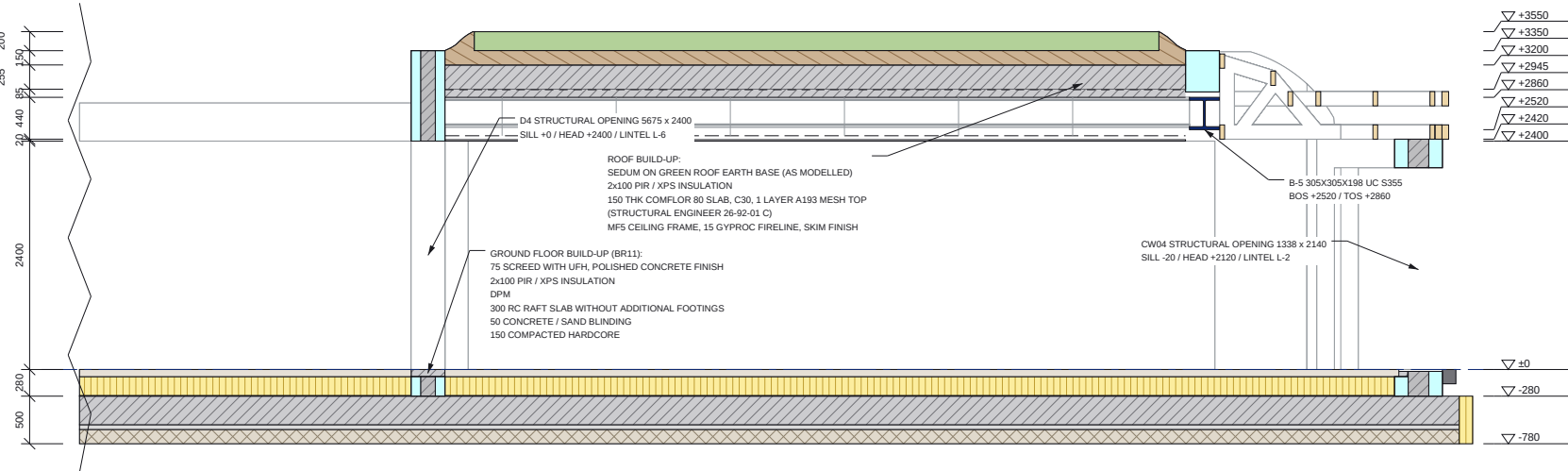
STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-SE-02B	P01

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION



SECTION C-C (PART 1 OF 2)

Scale 1:75 @ A3 | Looking north; levels above GF FFL



SECTION C-C (PART 2 OF 2)

Scale 1:75 @ A3 | Looking north; levels above GF FFL

- ICF EPS (cut)

ICF concrete core / RC (cut)

Steel (cut)

Screed
- Insulation (PIR / XPS, perimeter)

Blinding / hardcore

Roof earth

Sedum
- Masonry wall (cut)

Suspended ceiling (MF5 + 15 Fireline)

Comflor 80 deck (profile line)

Beyond the cut (line work)



SECTION C-C SCHEDULES

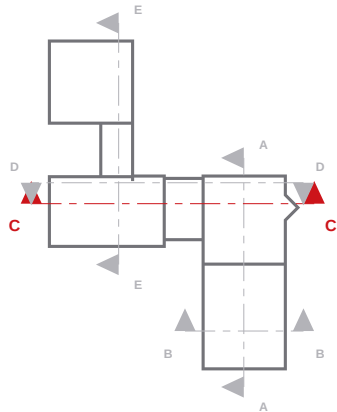
LEVELS ON THIS SECTION

Level	FFL	AOD
underside of hardcore	-780	142.970
underside of ICF / top of RC raft	-280	143.470
GF FFL	0	143.750
underside of suspended ceiling	+2400	146.150
top of ICF	+2420	146.170
B-5 BOS	+2520	146.270
roof slab soffit	+2850	146.600
underside of Comflor deck	+2860	146.610
roof slab soffit	+2945	146.695
top of roof slab	+3105	146.855
top of roof slab	+3200	146.950
top of ICF	+3350	147.100
top of roof build-up	+3454	147.204
top of roof build-up	+3550	147.300

STEEL CUT BY THIS SECTION

Mark	Section (engineer)	BOS	TOS
B-5	305x305x198 UC S355 with 200x100x10 ledger angle one side	+2520	+2860

SECTION KEY



NOTES

- Section C-C on the line of the architect's section (BR13). Cut elements hatched / filled; the model beyond the cut in line work.
- Levels above GF FFL (143.750 AOD); the level table gives AOD.
- ICF: Nudura 6" XR35, 101.6 EPS + 152.4 C30 core + 101.6 EPS = 355.6. EXT ICF150: 150mm core ICF walls, concrete grade C30, 10mm aggregate; polypropylene fibre reinforcement, dosage 2.5kg/m3; H16 dowel bars between lifts @ max 450mm c/c, 800mm long with 200mm embedment, resin fixed.
- Ground floor (BR11): 75 screed with UFH / 200 PIR-XPS / DPM / 300 RC raft slab without additional footings / 50 blinding / 150 compacted hardcore; 100 perimeter insulation outside the ICF. Thresholds: ICF cut to -75 under the screed, concrete infill, ACO drain (size assumed 100x150).
- Roof: 150 thk Comflor 80 slab, C30, 1 layer A193 mesh top (26-92-01 C); green roof build-up (earth + sedum) as modelled; waterproofing to specialist design. Timber roof over the east wall to specialist supplier design. Comflor 80 deck drawn to the architect's profile (80 deep, 300 pitch) on the bearing steel / walls; MF5 suspended ceiling, 15 Gyproc Fireline, FFL to underside 2400 (BR11-13), drawn only.
- Steel marks, BOS / TOS: Carb Zero 26-92-01 rev C. See UKPCS-002-GA-01B / EL-01B / EL-02B.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

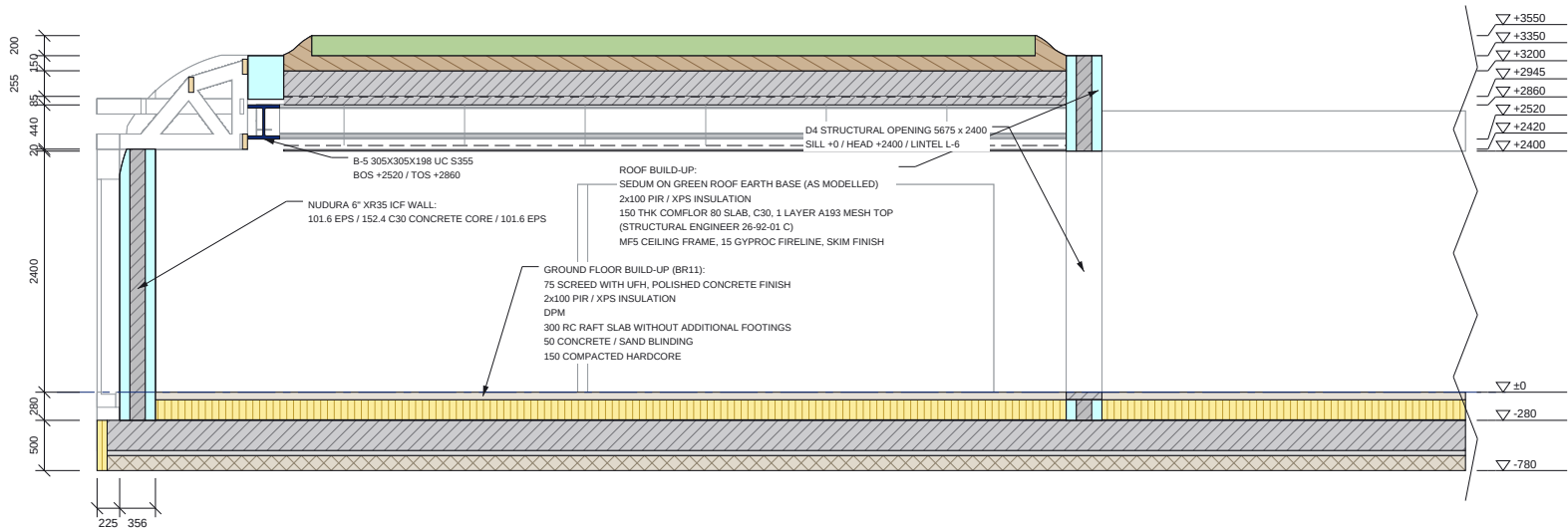
DRAWING TITLE
SECTION C-C

ICF section with levels and build-ups

SCALE	SIZE	DATE	DRAWN
1:75	A3	15.09.26	TS

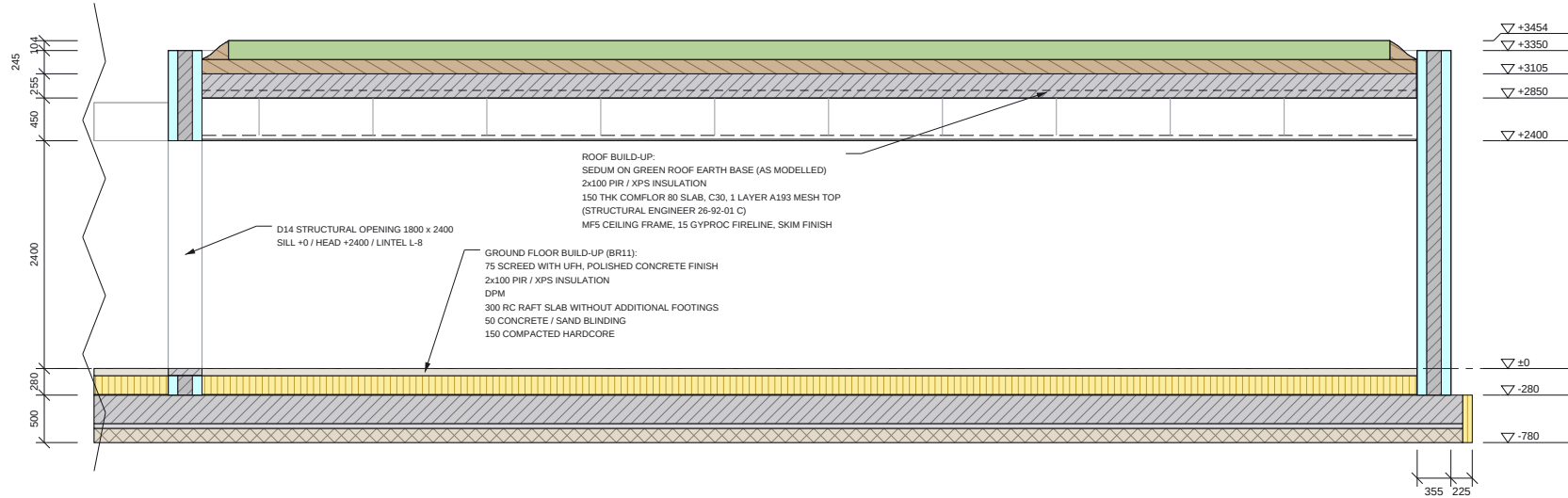
STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-SE-03B	P01

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION



SECTION D-D (PART 1 OF 2)

Scale 1:75 @ A3 | Looking south; levels above GF FFL



SECTION D-D (PART 2 OF 2)

Scale 1:75 @ A3 | Looking south; levels above GF FFL

- ICF EPS (cut)

ICF concrete core / RC (cut)

Steel (cut)

Screed
- Insulation (PIR / XPS, perimeter)

Blinding / hardcore

Roof earth

Sedum
- Masonry wall (cut)

Suspended ceiling (MF5 + 15 Fireline)

Comflor 80 deck (profile line)

Beyond the cut (line work)



SECTION D-D SCHEDULES

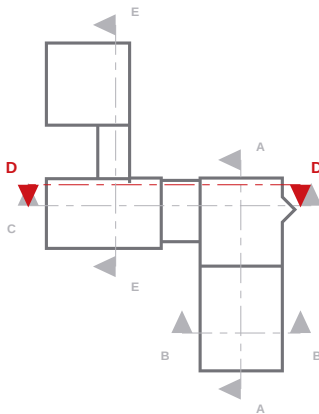
LEVELS ON THIS SECTION

Level	FFL	AOD
underside of hardcore	-780	142.970
underside of ICF / top of RC raft	-280	143.470
GF FFL	0	143.750
underside of suspended ceiling	+2400	146.150
top of ICF	+2420	146.170
B-5 BOS	+2520	146.270
roof slab soffit	+2850	146.600
underside of Comflor deck	+2860	146.610
roof slab soffit	+2945	146.695
top of roof slab	+3105	146.855
top of roof slab	+3200	146.950
top of ICF	+3350	147.100
top of roof build-up	+3454	147.204
top of roof build-up	+3550	147.300

STEEL CUT BY THIS SECTION

Mark	Section (engineer)	BOS	TOS
B-5	305x305x198 UC S355 with 200x100x10 ledger angle one side	+2520	+2860

SECTION KEY



NOTES

- Section D-D is a BuildEase ICF section (not an architect's section line). Cut elements hatched / filled; the model beyond the cut in line work.
- Levels above GF FFL (143.750 AOD); the level table gives AOD.
- ICF: Nudura 6" XR35, 101.6 EPS + 152.4 C30 core + 101.6 EPS = 355.6. EXT ICF150: 150mm core ICF walls, concrete grade C30, 10mm aggregate; polypropylene fibre reinforcement, dosage 2.5kg/m³; H16 dowel bars between lifts @ max 450mm c/c, 800mm long with 200mm embedment, resin fixed.
- Ground floor (BR11): 75 screed with UFH / 200 PIR-XPS / DPM / 300 RC raft slab without additional footings / 50 blinding / 150 compacted hardcore; 100 perimeter insulation outside the ICF. Thresholds: ICF cut to -75 under the screed, concrete infill, ACO drain (size assumed 100x150).
- Roof: 150 thk Comflor 80 slab, C30, 1 layer A193 mesh top (26-92-01 C); green roof build-up (earth + sedum) as modelled; waterproofing to specialist design. Timber roof over the east wall to specialist supplier design. Comflor 80 deck drawn to the architect's profile (80 deep, 300 pitch) on the bearing steel / walls; MF5 suspended ceiling, 15 Gyproc Fireline, FFL to underside 2400 (BR11-13), drawn only.
- Steel marks, BOS / TOS: Carb Zero 26-92-01 rev C. See UKPCS-002-GA-01B / EL-01B / EL-02B.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



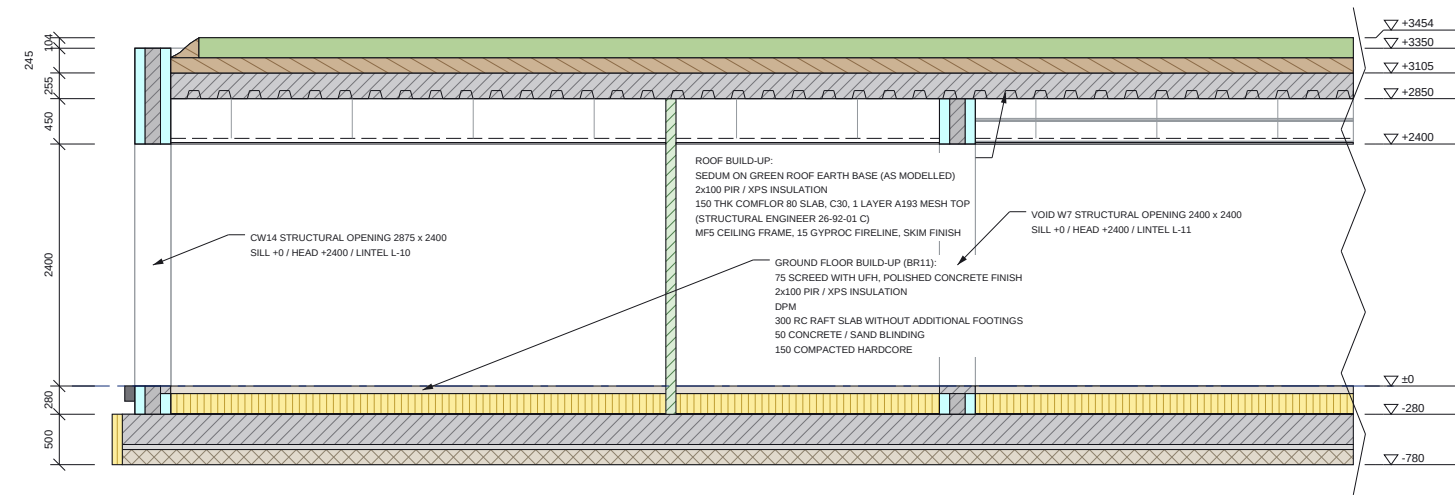
CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE
SECTION D-D
ICF section with levels and build-ups

SCALE	SIZE	DATE	DRAWN
1:75	A3	15.09.26	TS

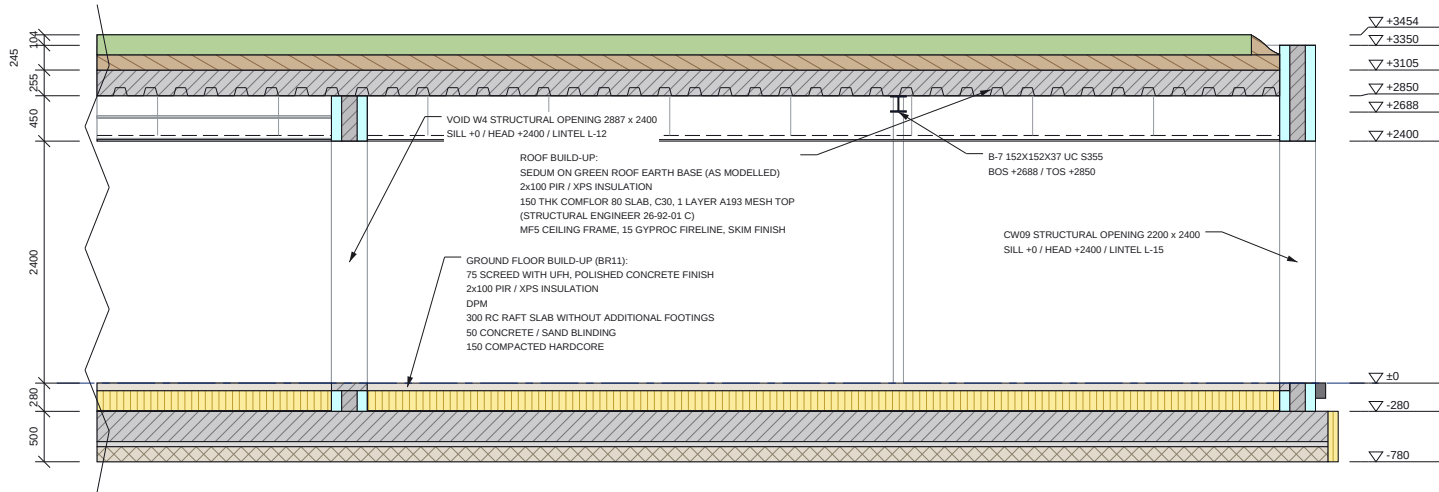
STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-SE-04B	P01

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION



SECTION E-E (PART 1 OF 2)

Scale 1:75 @ A3 | Looking west; levels above GF FFL



SECTION E-E (PART 2 OF 2)

Scale 1:75 @ A3 | Looking west; levels above GF FFL

- ICF EPS (cut)

ICF concrete core / RC (cut)

Steel (cut)

Screed
- Insulation (PIR / XPS, perimeter)

Blinding / hardcore

Roof earth

Sedum
- Masonry wall (cut)

Suspended ceiling (MF5 + 15 Fireline)

Comflor 80 deck (profile line)

Beyond the cut (line work)



SECTION E-E SCHEDULES

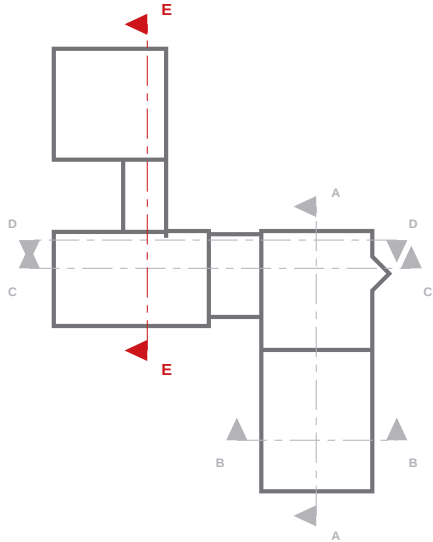
LEVELS ON THIS SECTION

Level	FFL	AOD
underside of hardcore	-780	142.970
underside of ICF / top of RC raft	-280	143.470
GF FFL	0	143.750
underside of suspended ceiling	+2400	146.150
B-7 BOS	+2688	146.438
top of ICF	+2850	146.600
top of roof slab	+3105	146.855
top of ICF	+3350	147.100
top of roof build-up	+3454	147.204

STEEL CUT BY THIS SECTION

Mark	Section (engineer)	BOS	TOS
B-7	152x152x37 UC S355 with 10mm bottom plate both sides	+2688	+2850

SECTION KEY



NOTES

- Section E-E is a BuildEase ICF section (not an architect's section line). Cut elements hatched / filled; the model beyond the cut in line work.
- Levels above GF FFL (143.750 AOD); the level table gives AOD.
- ICF: Nudura 6" XR35, 101.6 EPS + 152.4 C30 core + 101.6 EPS = 355.6. EXT ICF150: 150mm core ICF walls, concrete grade C30, 10mm aggregate; polypropylene fibre reinforcement, dosage 2.5kg/m3; H16 dowel bars between lifts @ max 450mm c/c, 800mm long with 200mm embedment, resin fixed.
- Ground floor (BR11): 75 screed with UFH / 200 PIR-XPS / DPM / 300 RC raft slab without additional footings / 50 blinding / 150 compacted hardcore; 100 perimeter insulation outside the ICF. Thresholds: ICF cut to -75 under the screed, concrete infill, ACO drain (size assumed 100x150).
- Roof: 150 thk Comflor 80 slab, C30, 1 layer A193 mesh top (26-92-01 C); green roof build-up (earth + sedum) as modelled; waterproofing to specialist design. Timber roof over the east wall to specialist supplier design. Comflor 80 deck drawn to the architect's profile (80 deep, 300 pitch) on the bearing steel / walls; MF5 suspended ceiling, 15 Gyproc Fireline, FFL to underside 2400 (BR11-13), drawn only.
- Steel marks, BOS / TOS: Carb Zero 26-92-01 rev C. See UKPCS-002-GA-01B / EL-01B / EL-02B.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE
SECTION E-E

SCALE	SIZE	DATE	DRAWN
1:75	A3	15.09.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-SE-05B	P01

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION

THRESHOLD DETAIL - EXTERNAL SCHEDULE

OPENINGS WITH THIS THRESHOLD

Ref	Width	Sill	Head	Lintel
CW04	1338	-20	+2120	L-2
CW05	4099	0	+2400	L-17
CW08	3250	0	+2400	L-14
CW09	2200	0	+2400	L-15
CW10	2200	0	+2200	L-15
CW13	2875	0	+2400	L-10
CW14	2875	0	+2400	L-10
CW01, D01, CW02	4098	0	+2400	L-16
D02	1250	0	+2400	L-3
D1	5775	0	+2400	-
D15	4250	0	+2060	L-1
D2	5125	0	+2400	L-13
D3	7855	0	+2400	L-7

Structural openings; sill and head above GF FFL. VOID = subtract cut in the ICF (no end caps).

LEVELS IN THIS DETAIL

Level	FFL	AOD
underside of hardcore	-780	142.970
underside of blinding	-630	143.120
underside of raft	-580	143.170
underside of ICF / top of RC raft	-280	143.470
DPM / top of raft	-275	143.475
underside of ACO	-150	143.600
ICF cut / top of insulation / threshold infill	-75	143.675
underside of bottom buck (external only)	-63	143.687
GF FFL / top of screed	0	143.750

NOTES

- Threshold detail cut square to the wall through D02; applies to every opening listed (same threshold type).
- External floor-level threshold: internal EPS leaf of the ICF cut to -75 (under the 75 screed), concrete infill -75 to the sill, ACO drain against the external face (size not given on the drawings: 100x150 assumed, opening width + 300 each side).
- Architect's threshold note (BR11 / BR12 / BR13): "Rebated curtain wall framing to suit level of access. PPC aluminium flashing to threshold. Below FFL ICF block 200mm high." Door / curtain wall frame, flashing and finishes by others, not shown.
- Ground floor build-up BR11. Levels above GF FFL (143.750 AOD). ICF Nudura 6" XR35; BCDS bottom buck / end cap as modelled.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE
THRESHOLD DETAIL - EXTERNAL

SCALE	SIZE	DATE	DRAWN
1:10	A3	15.09.26	TS

STATUS

FOR APPROVAL

DRAWING No.
UKPCS-002-DT-01B

REV
P01

THRESHOLD DETAIL - EXTERNAL (AT D02)

Scale 1:10 @ A3 | Section square to the wall through the opening; external side to the right



Do not scale. All dimensions in mm. Check all dimensions on site.

OPTION B - FOR APPROVAL - NOT FOR CONSTRUCTION

THRESHOLD DETAIL - INTERNAL SCHEDULE

OPENINGS WITH THIS THRESHOLD

Ref	Width	Sill	Head	Lintel
D13	1587	0	+2400	L-8
D14	1800	0	+2400	L-8
D4	5675	0	+2400	L-6
VOID W19	4143	0	+2070	L-5
VOID W4	2887	0	+2400	L-12
VOID W7	2400	0	+2400	L-11

Structural openings; sill and head above GF FFL. VOID = subtract cut in the ICF (no end caps).

LEVELS IN THIS DETAIL

Level	FFL	AOD
underside of hardcore	-780	142.970
underside of blinding	-630	143.120
underside of raft	-580	143.170
underside of ICF / top of RC raft	-280	143.470
DPM / top of raft	-275	143.475
ICF cut / top of insulation / threshold infill	-75	143.675
GF FFL / top of screed	0	143.750

NOTES

- Threshold detail cut square to the wall through D4; applies to every opening listed (same threshold type).
- Internal threshold between building sections: ICF cut full thickness to -75, full-depth concrete infill -75 to FFL, no ACO.
- Architect's threshold note (BR11 / BR12 / BR13): "Rebated curtain wall framing to suit level of access. PPC aluminium flashing to threshold. Below FFL ICF block 200mm high." Door / curtain wall frame, flashing and finishes by others, not shown.
- Ground floor build-up BR11. Levels above GF FFL (143.750 AOD). ICF Nudura 6" XR35; no lintel closure / bottom buck at internal thresholds.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE
THRESHOLD DETAIL - INTERNAL

SCALE	SIZE	DATE	DRAWN
1:10	A3	15.09.26	TS

STATUS

FOR APPROVAL

DRAWING No.

UKPCS-002-DT-02B

REV

P01

THRESHOLD DETAIL - INTERNAL (AT D4)

Scale 1:10 @ A3 | Section square to the wall through the opening



Do not scale. All dimensions in mm. Check all dimensions on site.

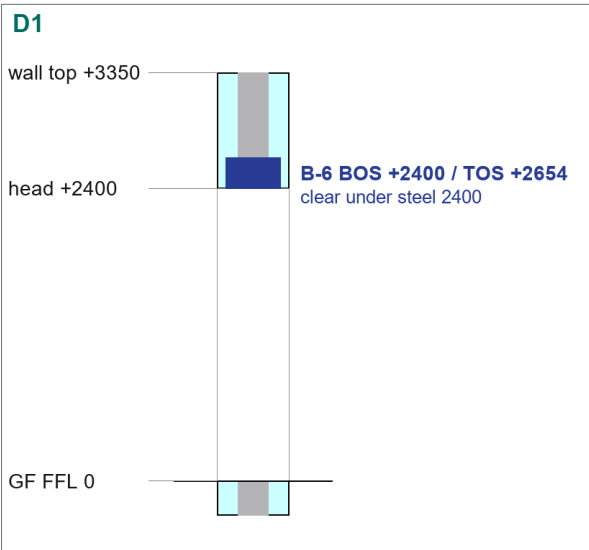
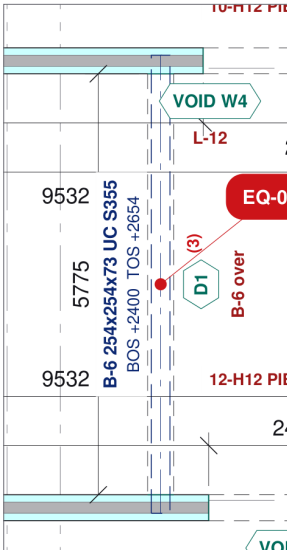
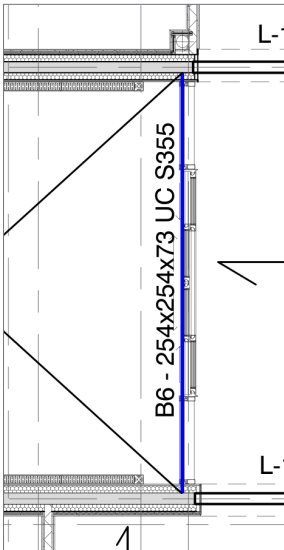
EQ-02 D1: steel B-6 as the head of the 5575 opening (confirm interpretation)

Status: Open | see GA-01

Engineer Carb Zero 26-92-01 rev C

ICF model (GA-01)

Section at D1 (model)



ISSUE D1 is the 5775 wide full-length opening in the ICF wall between the conservatory and the room to its east (Wall 21). Your drawing gives it no lintel mark and shows B-6 254x254x73 UC S355 (with 200x100x10 ledger angle one side) along that line. We have modelled B-6 at BOS +2400 / TOS +2654 as the head of the opening, with the structural opening in the ICF up to +2654 and ICF over the steel to the wall top at +3350, giving 2400 clear under the steel.

QUESTION Please confirm that B-6 forms the head of D1 as described (clear opening 2400 under B-6, ICF bearing on the steel above), or advise the intended arrangement.

ENGINEER'S RESPONSE

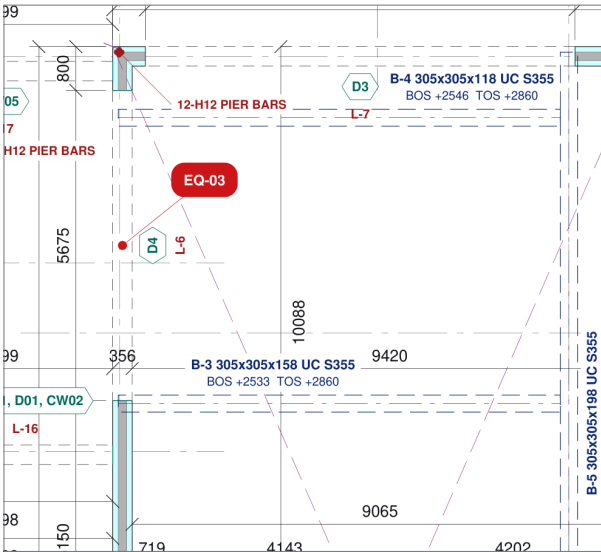
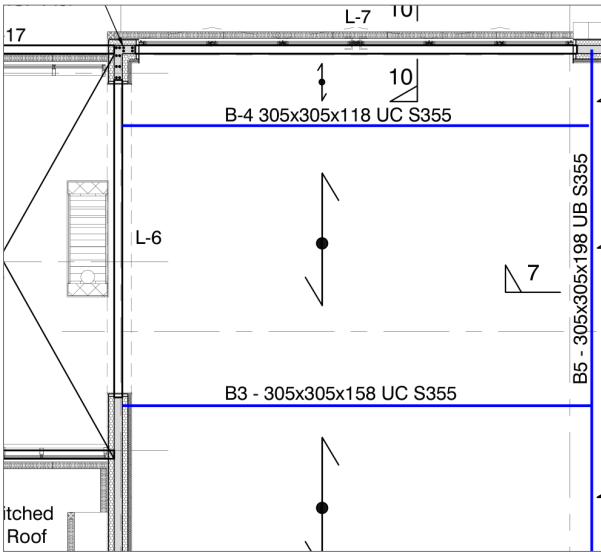
Name / date:

EQ-03 Option B: steel lengths and bearings with the external ICF moved 100mm in

Status: Open | see GA-01

Engineer Carb Zero 26-92-01 rev C

ICF model (GA-01)



ISSUE In Option B every external ICF wall moves 100mm in from the RC raft edge. Steels have been adjusted to keep their bearing into the ICF: B-3, B-4 (west ends) and B-5 (north end), B-1 (south end) and B-7 (east end) are 100mm shorter; B-9 (south end) and the tie (west end) are 100mm longer; B-6 moves 100mm east and is 200mm longer (D1 5775). B-2, B-8 and C-1 are unchanged. The engineer's pier bars and wall starters move with the ICF corners and junctions; lintel spans over VOID W4 / W7 reduce by 200mm, D1 / CW05 / CWO1 openings widen by 200mm.

QUESTION Please confirm the steel lengths / bearings, the 200mm longer B-6 and the relocated pier bars are acceptable for Option B, and whether the RC raft edge detail (ICF now ~125mm in from the slab edge) needs revising.

ENGINEER'S RESPONSE

Name / date:

QUESTIONS REGISTER

QUERIES

Ref	Subject	Drawing	Status
EQ-02	D1: steel B-6 as the head of the 5575 opening (confirm interpretation)	GA-01	Open
EQ-03	Option B: steel lengths and bearings with the external ICF moved 100mm in	GA-01	Open

NOTES

- Questions raised on the ICF approval drawings against the structural engineer's drawing Carb Zero 26-92-01 rev C.
- Each panel shows the engineer's drawing beside the architect's drawing and/or the ICF model, cropped to the same area.
- Please answer in the response box of each question (or by email quoting the EQ reference); the approval drawings will be revised to suit.
- EQ tags on the GA drawings link to these panels.

REVISIONS

REV	DATE	DESCRIPTION
P01	15.09.26	Approval issue - Option B: external ICF 100mm in from slab edge



CLIENT	PROJECT
Mr A Kirkpatrick	11767 Horrobin Farm, Turton

DRAWING TITLE
QUESTIONS FOR THE ENGINEER

Structural queries on the ICF approval drawings - sheet 1 of 1

SCALE	SIZE	DATE	DRAWN
NTS	A3	15.09.26	TS

STATUS	DRAWING No.	REV
FOR APPROVAL	UKPCS-002-EQ-01B	P01